

Supplementary Information

Agentic diagnosis of lower respiratory tract infection from metagenomic sequencing and multimodal clinical data

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Supplementary Methods

These sections give the detail summarized in the Methods of the main text.

Metagenomic sequencing workflow

mNGS was ordered by the treating physicians as part of clinical care and performed by one clinical laboratory (Asia Pathogenomics, New Taipei City, Taiwan) using a previously described workflow^{1,2}. Specimens were predominantly bronchoalveolar lavage fluid (60%) and blood (23%; Table 1 of the main text). Briefly, specimens were heat-inactivated and mechanically lysed. Nucleic acids were extracted without host depletion using the TIANMicrobe Pathogen DNA Kit (Tiangen) and, for RNA, the QIAamp Viral RNA Mini Kit (Qiagen)¹. Libraries were prepared with unique dual-index adapters (TIANSeq Direct Fast DNA Library Prep Kit, Tiangen) and sequenced as 75-bp single-end reads on an Element Biosciences AVITI system, yielding approximately 40 million reads per sample².

Reads were quality-filtered with Trimmomatic³ and PRINSEQ⁴. Human reads were removed by alignment to T2T-CHM13⁵ with BWA-MEM⁶, and the remaining reads were aligned to a curated database containing 5,884 viral, 18,432 bacterial, 3,146 fungal and 370 parasitic genomes². Extraction, library and no-template negative controls were included in each run. Organisms were reported when normalized abundance exceeded ten times the highest value in a concurrent negative control. For fastidious, low-biomass pathogens such as *Mycobacterium tuberculosis*, uniquely mapping reads were required in the specimen and absent from all controls². AETIA used only the resulting clinical report—the reported organisms, read counts and ranks in the negative-control comparison—and did not process raw sequencing reads.

Conventional microbiology

Conventional microbiology was performed in each hospital’s accredited clinical laboratory as part of routine care: culture with identification by mass spectrometry or automated biochemical systems; the BioFire FilmArray Pneumonia Panel (bioMérieux) on lower respiratory specimens, which reports typical bacteria semi-quantitatively and atypical bacteria, viruses and resistance genes qualitatively^{1,7}; the Platelia *Aspergillus* galactomannan enzyme immunoassay (Bio-Rad) in serum and lavage fluid⁸; and targeted molecular tests. Results were taken as reported. Within the pipeline, a FilmArray bacterial target contributes evidence in proportion to its semi-quantitative bin, and galactomannan is considered positive at an index of ≥ 1.0 in lavage fluid and ≥ 0.5 in serum, in line with consensus definitions⁹.

Literature evidence, case matching and the independent evidence dimensions

For every initial selection and additional candidate retained by the safety review, PubMed was searched through NCBI E-utilities using the organism name and lower-respiratory-tract infection terms. Searches included publications available by 30 June 2026 and returned up to ten articles per organism. A language model (GPT-5.6 Luna) assessed each title and abstract for human clinical evidence that the exact species causes infection at the relevant anatomical site. Supporting evidence required an exact-species match, human infection at the target site

and a traceable excerpt from the source. When too few articles could be assessed, the literature analysis abstained rather than treating limited evidence as evidence against infection (module A; Supplementary Table 4).

The same articles were then compared with a label-blind summary of the individual patient (module A1). This summary included specimen type and site, clinical syndrome, host factors, imaging and laboratory phenotype and timing, but excluded the reference label, initial selections, read counts, culture and PCR results and all previous review rationales. The model extracted the reported disease context and graded agreement in anatomical site, syndrome, specimen, host, phenotype and timing. Deterministic rules converted these structured observations to strong match, partial match, mismatch or insufficient evidence. Exact-species causal evidence with compatible site and syndrome was required for a positive match; explicit colonization, a conflicting site or syndrome, non-human evidence or broader taxonomic evidence could not support rescue. Only articles with traceable supporting or opposing excerpts entered the aggregate case-match score.

Three additional dimensions summarized evidence independent of article case matching (Supplementary Table 4). Sequencing strength captured both absolute burden and relative prominence within the specimen (module B). Direct microbiological evidence was graded by assay specificity and anatomical relevance, with exact-species detection in a sterile site or tissue receiving the strongest grade (module C). Results without a usable specimen site or with contamination concerns were treated as context rather than confirmation, and respiratory *Candida* culture alone was not considered evidence of invasive disease. A final check recorded anatomical coherence and colonization concerns for audit (module D). It did not determine selection, but an explicit anatomical mismatch could not be overridden.

Direct-prompting runs

Each model was run once per patient. The response, prompt hash, payload hash, requested and returned model identifiers, token usage and a UTC timestamp were stored before scoring. Requests that failed at the transport level were retried up to twice with exponential back-off, and the output schema was enforced by the provider. After the first runs of GPT-5.6 Sol, GPT-5.6 Luna and GPT-5.6 Terra, a correction to the linkage between patients and their input records changed the payload of 22 of the 41 patients. These 22 were re-queried with the unchanged prompt, model and settings; the stored responses of the other 19 were reused after verification of their prompt and payload hashes. GPT-6 Astra was first queried after the correction. No response was edited.

Secondary registry analyses

The four exploratory registry analyses summarized in the Methods are described below. None was used to develop or evaluate the pathogen-selection rules.

Concordance of sequencing and conventional microbiology

To characterize the information provided by the mNGS report before multimodal integration, we analysed registry cases with an adjudicated pathogen list in the shared workbook ($n = 33$; Supplementary Note 6). KMH cases were not included in this analysis because their adjudications were stored separately in per-patient workbooks. For each case, we compared the organisms and read counts in the mNGS report with organisms detected by culture and non-culture tests and with the adjudicated pathogen list. Names were normalized and matched at exact-species level using the same rules as the primary evaluation. Entries explicitly indicating that no pathogen was present were excluded.

An mNGS detection was considered corroborated when the same species was identified by any conventional test in the same patient. For adjudicated pathogens absent from the mNGS report, we recorded the detecting assay, specimen type and collection time relative to receipt of the mNGS specimen. Conventional assays included culture, PCR and multiplex panels, viral-load testing, antigen testing and IgM or antibody serology. Bronchoalveolar lavage fluid and sputum were classified as non-sterile specimens; blood, cerebrospinal fluid, tissue, ascites, pus and abscess material were classified as sterile. The ability of read count alone to distinguish causative from non-causative detections was summarized by the area under the receiver operating characteristic curve, computed for the 80 of 104 detections with a recorded read count.

Composition and timing of the clinical record

To quantify the evidence available for physician interpretation (Fig. 3a–d), we analysed the complete per-admission source workbooks from KMH. These records contained underlying conditions; admission, transfer and discharge diagnoses; culture reports with collection and reporting times; full-text radiology reports; all syndromic-panel targets; galactomannan measurements; and blood count and chemistry results. Unlike the registry summary, they retained negative results, timestamps and the original report wording.

Of 36 workbooks, 32 distinct admissions were analysed. One lacked microbiology, imaging and panel data; one was labelled as a duplicate; and two were byte-identical copies of another admission that had undergone repeat mNGS. Repeat sequencing of the same admission was counted once. We defined a result as one culture report, radiology study, panel target, galactomannan measurement or recorded analyte value. Organisms were counted once per admission and read-out after synonym normalization; galactomannan was considered to identify *Aspergillus* at an index of ≥ 0.5 in serum or ≥ 1.0 in lavage, and resistance markers were not counted as organisms. Sterile-site cultures comprised blood, ascites, pleural fluid, tissue, cerebrospinal fluid, abscess, deep-wound and intravascular-catheter specimens. Culture turnaround was the interval from specimen collection to the final report.

Radiology wording was classified as qualified when a report used terms such as “probably”, “suspect”, “favour”, “cannot be excluded”, “questionable”, “equivocal”, “may represent” or “suggestive of”. Formulaic recommendations for clinical correlation or differential diagnosis were not considered qualifications. Only aggregate results from these protected source records were exported for analysis.

For each read-out, we summarized the interval between specimen collection and receipt of

the mNGS specimen, the interval between admission and mNGS collection and the proportion of patients in whom that read-out identified an organism. Collection intervals within 90 days were summarized by the median, interquartile range and 5th–95th centiles; larger intervals were treated as date-entry errors. Because the registry records conventional tests almost exclusively when an organism was detected (225 of 226 dated culture entries were positive), these distributions describe the timing of organism-identifying evidence, not the complete testing burden. They therefore cannot estimate culture yield or the effect of previous antimicrobial treatment.

The registry recorded arrival and reporting dates for mNGS specimens, but these dates were identical for more than half of the records and were not used to estimate sequencing turnaround. These analyses were descriptive, and no inferential test was applied. Because adjudicators had access to the mNGS report, comparisons with the reference standard share the same non-independence as the primary evaluation. Only de-identified per-detection data and aggregate counts were exported from the protected registry.

Physiological trajectory around the sequencing report

We descriptively examined physiological trajectories around mNGS specimen collection using machine-readable extracts of the complete KMHU discharge record. Each dated laboratory value and vital sign retained a link to its source page. Extracts were available for 49 of 51 KMHU registry admissions; 48 also had a dated mNGS specimen and recorded discharge status and were included (31 deaths and 17 survivors).

The collection date was defined as day 0. Measurements were grouped into five windows: days -14 to -1 , $0-2$, $3-7$, $8-14$ and $15-30$. For each patient and window, we first calculated the median value of each marker so that patients with frequent measurements contributed once per interval; group summaries were then calculated from these patient-level medians. Pre-collection and day $3-7$ values were compared with a two-sided Wilcoxon signed-rank test among patients with measurements in both windows, without adjustment for multiple comparisons. Fever was summarized as the maximum temperature within each window and defined as a value above 38°C . Improvement was defined as movement in the clinically favourable direction between the pre-collection and day $3-7$ windows; improvement followed by worsening required a subsequent reversal during days $8-30$.

Values outside prespecified physiological ranges were excluded. Temperature values were re-constructed from the source table layout and underwent a documented correction for a systematic transcription error, which was checked against adjacent readings from the same admission.

This analysis was observational and descriptive. All patients underwent mNGS, and there was no unexposed comparator. Antipyretics, corticosteroids, sedation and ventilator settings could directly affect temperature and gas exchange but were not available in a form suitable for adjustment. Antimicrobial start and stop dates were also unavailable; therefore, neither time from reporting to treatment modification nor coverage of the adjudicated pathogen could be determined. The analysis was not used to make causal claims or to propose a monitoring test.

Registry treatment and outcome analysis

Treatment modification, its direction, its attribution to mNGS, mechanical ventilation and discharge status were extracted for all 105 registry specimens and summarized as counts and percentages. Denominators were reported when attribution or ventilation status was missing. The number of organisms per mNGS report was compared across specimen types and patient strata. Because specimen type was associated with disease severity, stratum comparisons were repeated within bronchoalveolar-lavage specimens, and only these restricted comparisons were interpreted. Organism-level agreement with culture was calculated after the same name normalization used in the primary evaluation and cross-checked against the registry’s existing comparison.

Because every registry patient underwent mNGS, the data could not estimate the effect of sequencing relative to no sequencing. We therefore compared patients whose treatment modification was retrospectively attributed by the treating team to mNGS with those whose modification was not. This attribution was neither assigned nor assessed under blinding. In-hospital mortality was compared with Fisher’s exact test. Odds ratios and 95% confidence intervals used the Haldane–Anscombe correction and were additionally estimated with logistic regression adjusted for intensive care at sampling, acute respiratory distress syndrome, active malignancy, corticosteroid treatment and age; Wald intervals are reported.

Comparisons between patients whose treatment changed for any reason and those whose treatment did not were reported descriptively because of evident confounding by indication. The registry did not contain antimicrobial names or administration dates, precluding assessment of treatment appropriateness or time to appropriate therapy. Time-to-event outcomes were also not analysed because patients had to survive long enough to receive a change attributed to a specimen collected after admission. All tests were two-sided, without adjustment for multiple comparisons. These secondary analyses were considered hypothesis-generating, and only aggregate results were exported from the protected registry.

Supplementary Tables

Table S1: **mNGS organism features computed before scoring.** Source: `upstream/tools/rank_mngs_candidate_microbes.py`, `upstream/tools/mngs_common.py`, `deterministic_mngs_max_scorer.py` (`dominance_by_record`).

Feature	Values	Definition
Control-subtracted priority	1–5	Rank of each reported organism in the laboratory’s negative-control comparison (1, no signal in any negative control; 5, weakest), as output with the mNGS report. Code score 1.0, 0.85, 0.65, 0.45, 0.30.
Read percentile	0–1	Percentile of $\log_{10}(\text{reads} + 1)$ among all organisms in the run.
Final score / label	high, medium, low	$0.6 \times \text{code score} + 0.4 \times \text{percentile}$; high ≥ 0.70 , medium ≥ 0.45 .
Read tier	R0–R4	R0: 1–9 reads; R1: ≥ 10 ; R2: ≥ 100 ; R3: $\geq 1,000$; R4: $\geq 10,000$.
Dominance tier	D0–D3	Per specimen record; top organism: D3 if reads $\geq 10 \times$ runner-up, D2 if $\geq 3 \times$, else D1; every other organism D0.
Specimen alignment	aligned, sterile/systemic, non-aligned	From the mNGS-to-specimen assignment and the target site (lower respiratory tract).
Hospital support	per module	Same-organism support from culture, FilmArray/galactomannan, molecular microbiology; “Support”, “Not support” or “Unknown”.
Host context	V0–V3, O0–O2	Host-vulnerability tier and opportunistic-coverage level; host expansion when V2–V3 or O1–O2.

Table S2: **Signal tiers and causative levels of the deterministic scorer (v20).** Source: `deterministic_mngs_max_scorer.py`, functions `m1_strong`, `m2_moderate`, `m3_protected`, `m4_weak`, `set_candidate_level_from_tier`, `level3_pickable`, `pickable`. Conditions are evaluated in order; the first satisfied tier applies.

Tier / level	Condition
M1 strong	priority 1–2 and R3–R4; or priority 1–2 and percentile ≥ 0.85 ; or D3 and $\geq R2$ and priority ≤ 3 .
M2 moderate	priority 1–2 and R2 and percentile ≥ 0.75 ; or D2–D3 and $\geq R2$ and priority ≤ 3 ; or same-organism auxiliary support and $\geq R1$ and priority ≤ 3 ; or protected pathogen (not reactivation virus or mould), $\geq R1$, priority ≤ 3 , label not low; or R3–R4 and priority ≤ 4 and label not low; or <i>P. jirovecii</i> with R1–R2 and priority ≤ 3 ; or Mucorales at priority 1 with percentile ≥ 0.60 . Reactivation viruses and <i>Aspergillus</i> use dedicated clauses.
M3 protected	retention condition of a protected pathogen; or priority 1–2 with R0–R2; or priority ≤ 3 with percentile ≥ 0.60 above R0; or priority ≤ 3 with high/medium label and R1–R2; or priority 3 with R1–R2 and non-host auxiliary support; or priority 4 with R4.
M4 weak	priority 4–5; or label low; or R0–R1; or non-aligned specimen.
M5 background	organism on the background list.
Level 1	M1, aligned or sterile specimen, and (D3 or R3–R4).
Level 2	other M1; M2 with aligned or sterile specimen.
Level 3	M2 non-aligned; M3; M4 under host expansion when plausible (not reactivation virus, colonizer or non-aligned; with non-host support, protected status, or priority ≤ 3 with R1 and high/medium label).
Level 4	other M4; any organism demoted by a guardrail.
Level 5	M5.
Picked	level 1–2; level 3 with protected status, an oral-flora/respiratory-commensal representative rule, related typical-pneumonia context, direct support from culture, FilmArray/galactomannan or molecular testing, or (host expansion) priority ≤ 2 with $\geq R2$ or D2–D3; <i>Candida</i> /generic yeast never at level 3 without invasive support. Fallback: level-4 “best available” when nothing is pickable.

Table S3: **Guardrail rule identifiers of the deterministic scorer.** All 33 R-S4-* identifiers in `deterministic_mngs_max_scorer.py`. Descriptions summarize the identifier and rule body.

Identifier (R-S4-...)	Effect
BACKGROUND	background organism, level 5
CANDIDA-BLOOD-STERILE-CONTEXT	<i>Candida</i> retained on blood/sterile-site evidence
CANDIDA-HOSPITAL-NONINVASIVE-NOT-PICKED	<i>Candida</i> with only non-invasive hospital support not picked
CANDIDA-RESP-CONTEXT-NOT-PICKED	respiratory <i>Candida</i> context not picked
CANDIDA-RESP-L3	respiratory <i>Candida</i> held at level 3; picked only with invasive support
COMMON-HOSPITAL-R0R1-NONDOMINANT-PICKED-OMIT	common hospital organism with R0–R1, non-dominant: omitted from picks
GENERIC-UNIDENTIFIED-HOSPITAL-LABEL	generic labels (e.g. “Gram-negative bacilli”) never picked
HERPES-HOST-R2R3-NONDOMINANT-REVIEW	herpesvirus with host-only support, R2–R3, non-dominant: review only
HERPES-R4-NONDOMINANT-HOST-ONLY-REVIEW	herpesvirus with R4 but non-dominant and host-only: review only
IMPOSSIBLE-INFECTION-SOURCE	organism cannot be a pulmonary infection source
L3-MNGS-ONLY-HOST-LOW-NONDOMINANT	level-3 mNGS-only organism with low, non-dominant reads not picked
LOW-ACTIONABILITY-VIRUS-PICKED-OMIT	EBV, HHV-6/7 and similar omitted from picks
LOW-PULM-SPECIFICITY-ATYPICAL-L4	atypical organism of low pulmonary specificity, level 4
LOW-PULM-SPECIFICITY-ENV-L4	environmental organism of low pulmonary specificity, level 4
LOW-SPECIFICITY-VIRUS-L4	low-specificity virus, level 4
MNGS-ONLY-NONDOMINANT-NON-TYPICAL-PICKED-OMIT	mNGS-only, non-dominant, non-typical pneumonia organism omitted
MNGS-R0R1-HOST-ONLY-NONPROTECTED-PICKED-OMIT	R0–R1 with host-only support, non-protected: omitted
NON-CORE-RESP-VIRUS-PICKED-OMIT	non-core respiratory virus omitted from picks
NONPULMONARY-CULTURE-CONTEXT-ONLY	culture at a non-pulmonary site is context only
ORAL-ASPIRATION-FLORA-L3	oral/aspiration flora representative retained at level 3
ORAL-ASPIRATION-FLORA-MNGS-ONLY-NONDOMINANT-L4	mNGS-only, non-dominant oral flora, level 4
RARE-YEAST-HOSPITAL-MNGS-L3	rare opportunistic yeast with hospital and mNGS support, level 3
RARE-YEAST-MNGS-ONLY-CONTEXT	rare yeast with mNGS only: context
RELATED-REPRESENTATIVE-HOSPITAL-CONTEXT	related-organism hospital context supports a representative
RESP-COMMENSAL-L3	respiratory commensal retained at level 3
RESP-COMMENSAL-L4	respiratory commensal, level 4
SAME-REPRESENTATIVE-GROUP-PICKED-OMIT	one representative per taxonomic group; others omitted
SKIN-ENV-YEAST-L4	skin/environmental yeast, level 4
SKIN-LOW-SPECIFICITY-BACTERIA-L4	skin flora of low specificity, level 4
TRACE-PROTECTED-WATCHLIST-PREFERRED	trace-read protected pathogen kept on a watch-list, not picked
WATER-ENV-GNB-L4	water-borne environmental Gram-negative, level 4
WATER-ENV-LOW-SPECIFICITY-PICKED-OMIT	water-borne organism of low specificity omitted from picks
WEAK-MOLD-HOST-ONLY-PICKED-OMIT	weak mould signal with host-only support omitted

Table S4: **Evidence-module thresholds used by R5.** Sources:
RAG_re/config/default.json, RAG_re/config/formal_merge_full_a.json,
RAG_re_casefit/config/default.json, RAG_re_clinical/config/clinical_v1.json,
OBER_patient_delivery/generate_r5_final_decisions.py.

Module	Setting
A retrieval	PubMed E-utilities; query "organism"[Title/Abstract] AND (site terms in title/abstract); publication cutoff 2026-06-30; at most 10 articles; 0.34 s between requests; literature also retrieved for upstream picks (<code>literature_on_baseline_picked = true</code>).
A judgement	support requires exact species, human clinical evidence, target-site match and a valid evidence span (≤ 20 contiguous words); positive when ≥ 3 valid supports among ≥ 5 judgeable articles; otherwise abstain.
A1 case-fit	per-article verdict strong / partial / mismatch / insufficient (Methods); aggregation thresholds in config: ≥ 2 strong for high, ≥ 1 strong or ≥ 3 partial for moderate, ≥ 5 judgeable articles. R5 uses strong ≥ 2 or strong+partial ≥ 7 .
B	absolute axis: M1/M2 or R3/R4; relative axis: dominance $\geq D2$, or priority ≤ 1 with percentile ≥ 0.8 ; site alignment required; the optional negative-control-ratio and reads-per-million gates are disabled (<code>require_ntc_ratio = false</code> , <code>require_rpm = false</code>).
C	positive modules: culture, FilmArray, FilmArray/galactomannan, targeted molecular, molecular microbiology, PCR, antigen, galactomannan; grades C3 / C2 / C1 / C0 / CNEG as in Methods; respiratory <i>Candida</i> with culture only $\rightarrow$ C1.
D	pass / guarded / block / unknown; reported, not gated.
Temporal	window 3 days before and after the anchor (research arms only).
R5	picked OR C3 OR ((A1_strong ≥ 2 OR A1_match ≥ 7) AND site_aligned AND (B_absolute OR B_relative)); rule id R5_A1_UNION_AND_B_LOOSE.

Table S5: **Language models, prompts and configurations by stage.** SHA-256 prefixes of the files in the code repository at the time of writing (2026-09-20).

Stage	Model / setting	File	SHA-256 (prefix)
Record standardization (raw sections → standardized JSON)	GPT-5, temperature 1	upstream/core/llm_parser.py	–
Slide-image transcription (image input)	GPT-5.6 Luna, Responses API, image detail high	reference_versions/shared_upstream/ppt_patient_parser/vision_hooks.py	–
Host/laboratory agent	GPT-5	agents/prompts/CBC_Lab_underlying_prompt.txt	0929360cfb5d
Culture agent	GPT-5	agents/prompts/culture_agent_prompt.txt	e1631735b119
FilmArray and GM agent	GPT-5	agents/prompts/filmarray_GMtest_prompt.txt	2a092516ff13
Imaging agent	GPT-5	agents/prompts/image_agent_prompt.txt	b0ea51281d8f
Molecular agent	GPT-5	agents/prompts/molecular_microbiology_prompt.txt	802ccf52ac71
mNGS-to-specimen	GPT-5 or deterministic v4	agents/prompts/mNGS_to_specimen_prompt.txt	643dcf157308
Deterministic summary	rules v3 (reported) / v4 (current)	tools/build_deterministic_summary.py; build_deterministic_summary_v3_recovered.py	–
Scorer	rules v19 (reported) / v20 (current)	tools/deterministic_mngs_max_scorer.py	–
Safety review of unpicked organisms	GPT-5, reasoning medium	prompt in tools/review_missed_mngs_candidates.py	–
Post-review rules	deterministic: name canonicalization, weak-environmental omission, non-pulmonary-support guardrail, syndrome-pattern and tier convergence, atypical-serology rescue, group representatives	tools/merge_deterministic_max_with_missed_review.py	–
Article judge (A)	GPT-5.6 Luna, reasoning low, 1,200 tokens	RAG_re/prompts/article_support_v1.txt; config/default.json; formal_merge_full_a.json	af4178902c50; bc123c20c743; b3ca56742dce
Case-fit (A1)	GPT-5.6 Luna, reasoning low, 1,400 tokens	RAG_re_casefit/prompts/article_casefit_v1.txt; evidence_span_repair_v1.txt; config/default.json	e495aef60640; f6f15feca9d9; a9f387850b01
Clinical rules (B, C, D)	clinical_v1	RAG_re_clinical/config/clinical_v1.json	41121351788c
R5	rule R5_A1_UNION_AND_B_LOOSE	OBER_patient_delivery/generate_r5_final_decisions.py	–
Rationale	GPT-5.6 Luna, reasoning low	generate_llm_rationales.py	–
Direct prompting	gpt-5.6-sol/luna/terra, gpt-6-astra; reasoning medium; 4,000 tokens	LLM_test/direct_raw_benchmark/prompt_v1.txt	961087b8acdf

Supplementary Notes

Supplementary Note 1. Missed-candidate safety-review prompt

The review prompt is embedded in `upstream/tools/review_missed_mngs_candidates.py` (`build_prompt`). The model receives the review queue, deterministic output, hospital summary and mNGS specimen assignments and returns one of four categories: `review_high_priority`, `review_context_needed`, `review_low_specificity` or `omitted_with_reason`. It may not replace deterministic selections or upgrade an organism because of immunocompromise or protected status alone. Respiratory *Candida* and generic yeasts are considered low-specificity without same-species invasive evidence, and different *Candida* species are not treated as aliases. CMV, HSV and VZV are classified as reactivation without evidence of primary pulmonary disease; EBV is not considered a pulmonary candidate. Additional rules address non-dominant herpesvirus signals, oral and aspiration flora, water-borne Gram-negative bacteria, gastrointestinal pathogens, coagulase-negative staphylococci and corynebacteria. The output is a JSON object with Traditional-Chinese rationale fields.

Supplementary Note 2. Direct-prompting baseline prompt

File `LLM_test/direct_raw_benchmark/prompt_v1.txt` (SHA-256 prefix 961087b8acdf), original in Traditional Chinese. English rendering:

You are an infectious-diseases clinical reviewer. Using only the single patient's data provided here, decide which microorganisms are most likely the clinical causative pathogens of the current pulmonary or lower-respiratory-tract infection episode.

This is the Direct-Raw baseline: do not use external search, PubMed, RAG, AETIA, any candidate pool or any earlier model result. The raw mNGS list may contain contamination, colonization, background organisms, low-level detections and organisms unrelated to the site of infection; "detected" does not mean "pathogen". Judge from the mNGS reads and the laboratory's control ranking, the specimen site, cultures, molecular tests, imaging, host status and timing. Respiratory *Candida*, low-specificity environmental organisms, skin flora and latent or reactivated viruses must not be listed as the infection source merely because they were detected. A positive result from a non-pulmonary specimen must not be treated as a pulmonary source without a pulmonary link. An empty array means "no record available", not a negative test. You may select zero, one or several pathogens; do not force an answer. Put only the organisms you finally judge to be the infection source in `predicted_pathogens`; if evidence is insufficient, put them in `excluded_detected_organisms` or `limitations`. Name pathogens at species level; do not merge different *Candida* or other species. Output must follow the given JSON schema; supporting evidence, counter-evidence, reason and limitations should be short and auditable, and hidden reasoning must not be output.

Supplementary Note 3. Case examples

The cases in Fig. 3e,f originated from the 2026-01-17 presentation of the first pipeline version in a KMH intensive-care series. In case 1, *Klebsiella pneumoniae* was dominant in the bronchoalveolar specimen (68,569 reads) and was selected in the presence of new bilateral consolidation, pleural effusions, increasing C-reactive protein and worsening hypoxaemia. *Candida albicans*

from the same specimen was classified as airway colonization because of low, non-dominant reads, repeated low-burden respiratory culture and the respiratory-*Candida* guardrail. In case 2, *Candida tropicalis* met the C3 direct-evidence criterion because several blood-culture sets grew the same species and the bronchoalveolar read burden was high.

The full record delivered for each organism contains: the disposition (selected or not) and the rule path that produced it; the value of modules A, A1, B, C and D; the per-modality findings recorded as support or opposition; the PubMed identifiers and judged spans of the case-matched articles; the specimen, site and timing used; and the SHA-256 hashes of the input files. [Replace these prototype vignettes with the delivered R5 records (selected organisms, rule paths, module results, audited clinical rationale, reference labels) and obtain approval for case-level presentation.]

Supplementary Note 4. Version boundaries and replay

The 55 reported R5 decisions used rule version v19 (`mngs_deterministic_level_only_v19_low_specificity_taxonomy`). The recovered v19 scorer reproduced all saved scorer outputs from the stored summaries and ranked mNGS records. The current scorer, v20 (`mngs_deterministic_level_only_v20_central_taxonomy_profile`), uses a central rule-based organism classifier, removes a retired language-model scoring branch and retains M1 for typical pneumonia bacteria with lower-respiratory culture or FilmArray support where v19 assigned M2. Replaying the same inputs reproduced all candidate names, selected sets, merge-stage selections and review-tier assignments. Candidate levels changed in 15 patients and selection order changed in five. The clinical-summary replay used current patient directories rather than a verified frozen agent snapshot; reconstructed hospital evidence differed from the stored summaries for three patients, without changing any reported selection. Sources: `docs/UPSTREAM_REPLAY_SUMMARY.json`, `docs/UPSTREAM_RECOVERY.md` and `docs/VERSIONS.md`.

Supplementary Note 5. Evaluation outputs to be added

[Per-patient tables (predicted and reference sets per method), per-hospital metrics, the 9 rescued organisms with correctness, error analysis of false negatives and false positives, confidence intervals.]

Supplementary Note 6. Registry analysis of the sequencing report alone

The shared registry contained adjudicated pathogen lists for 33 cases (Methods; `tables/make_discordance.py`; `figures-source/discordance.csv`). mNGS reported 104 organisms, of which 38 (37%) were adjudicated as causative and 66 (63%) as colonizing, contaminating or incidental. The read-count distributions overlapped across five orders of magnitude. Read count ranked a causative organism above a non-causative organism in 61% of pairs (AUC 0.61), with AUCs of 0.73 for bronchoalveolar lavage fluid and sputum and 0.56 for blood, cerebrospinal fluid and tissue. The largest count, 189,467 reads of *Acinetobacter nosocomialis* in lavage fluid, was adjudicated as colonization despite culture growth. *Pneumocystis jirovecii* with 7 reads in another lavage specimen was adjudicated as causative on the basis of a positive

lavage PCR.

Ten species were causative in one patient and non-causative in another: cytomegalovirus, Epstein–Barr virus, herpes simplex virus 1, *P. jirovecii*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *A. nosocomialis*, *Burkholderia cenocepacia*, *Candida albicans* and *Candida tropicalis*. The list includes airway colonizers and recognized pathogens from multiple organism classes. Species identity and a uniform read threshold did not replace patient-specific clinical context, particularly for non-sterile respiratory specimens.

Evidence from other modalities improved discrimination. An exact-organism conventional test corroborated 47% of causative detections and 18% of non-causative detections. Seventeen of 55 adjudicated pathogens (31%) were absent from the mNGS report of the sequenced specimen; 16 (94%) were identified by another test, including culture in eight, PCR in five, serum galactomannan in two and IgM serology in one ([figures-source/missed_by_mngs.csv](#)). Fourteen of these tests were collected within two days of the mNGS specimen, and the longest interval was five days. Eleven pathogens were identified in a different specimen type, and five were identified in the same specimen type by a different testing principle. These findings indicate complementary failure modes of sequencing and conventional tests. [This analysis covers the registry cases whose adjudication is recorded in the shared workbook; the KMUH line keeps its adjudication in the per-patient workbooks and is not included. Recompute over all 105 specimens once those labels are consolidated.]

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